FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Tuberculosis in Children

Tutorial | Week 3 · Fits, TB, kidneys & nutrition | Paediatrics | Dr Mawanda

Childhood TB is usually diagnosed without a positive smear, by putting together a TB contact, persistent symptoms (cough or fever over 2 weeks, poor weight gain, lethargy), examination, chest X-ray and Xpert Ultra on sputum, gastric aspirate or stool; drug-susceptible TB is treated with weight-band dispersible fixed-dose combinations (4 months for non-severe disease, 6 months standard, 12 months for TB meningitis and bone TB), and every child contact without TB gets preventive therapy.

Must know before the ward round

1. Suspect TB (UCG 2023) in any child with cough >2 weeks, fever >2 weeks, poor weight gain or weight loss for a month, reduced playfulness/lethargy, or a close (household) contact with pulmonary TB.
2. Children under 5 and HIV-positive children are most likely to progress from infection to disease, and to severe forms (TB meningitis, miliary TB); BCG at birth protects against these severe forms.
3. Bacteriological confirmation: Xpert MTB/RIF Ultra on sputum (expectorated or induced), gastric aspirate, nasopharyngeal aspirate or STOOL (WHO 2022); urine LF-LAM for HIV-positive children who are seriously ill or have advanced HIV. Most children are diagnosed clinically: a negative Xpert does not exclude TB.
4. WHO 2022 treatment decision algorithms (children <10 y with presumptive pulmonary TB): score symptoms, signs and CXR; a score above 10 means treat. The older Keith Edwards score (≥7 = treat) is still taught in Uganda.
5. Doses (WHO/UCG): isoniazid 10 mg/kg (7–15), rifampicin 15 mg/kg (10–20), pyrazinamide 35 mg/kg (30–40), ethambutol 20 mg/kg (15–25), plus pyridoxine; given as dispersible FDCs RHZ 75/50/150 and RH 75/50 by weight band: 4–7.9 kg 1 tab, 8–11.9 kg 2, 12–15.9 kg 3, 16–24.9 kg 4; ≥25 kg adult FDCs.
6. Regimens: standard 2RHZE/4RH (6 months); non-severe TB in children 3 months–16 years 2HRZ(E)/2HR (4 months; SHINE trial; UCG 2023 at general hospital level and above); TB meningitis and osteoarticular TB 2RHZE/10RH (12 months) or WHO's 6-month intensive 6HRZ-Eto for TBM.
7. TB meningitis: subacute fever, irritability, vomiting, headache, cranial nerve palsy, reduced consciousness; CSF lymphocytes, high protein, low glucose; add prednisolone 2 mg/kg/day (max 60 mg) for 4 weeks, then taper over 2 weeks.
8. TPT: every household contact under 5 years (and every HIV-positive contact) once active TB is excluded by symptom screen. Options: 3HP (weekly isoniazid + rifapentine for 12 weeks, ≥2 years), 3HR (daily RH for 3 months, any age), 6H (isoniazid 10 mg/kg daily for 6 months), 1HP (≥13 y). Contacts of MDR-TB: WHO 2024 recommends 6 months of levofloxacin (UCG 2023 says no TPT).
9. TB/HIV: start TB treatment first, then ART within 2 weeks (delay 4–8 weeks in TB meningitis); with rifampicin, give dolutegravir twice daily (and for 2 weeks after rifampicin stops); super-boost LPV/r; cotrimoxazole prophylaxis for all.
10. Weigh every month and move up a weight band when the child gains weight; under-dosing is a common cause of poor response.
11. Children with TB are rarely infectious, but their diagnosis means an adult source case is still coughing at home: always trace the source.
12. TB is the leading single infectious killer (WHO); in 2024 an estimated 1.2 million children fell ill with TB worldwide (WHO fact sheet 2026).

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1. Definition

Tuberculosis (TB) in children is infection and disease caused by *Mycobacterium tuberculosis*, almost always caught from an **adult or adolescent with pulmonary TB in the same household**. Children differ from adults in three ways that shape the whole tutorial:

1. **They progress quickly and severely**: infants and under-fives, HIV-positive and malnourished children are the most likely to develop disease, including **TB meningitis** and **miliary TB**, often within a year of infection.
2. **They are hard to confirm**: they rarely produce sputum and have few bacilli (paucibacillary), so most are **diagnosed clinically**, supported by a contact history, a chest X-ray and Xpert Ultra on **sputum, gastric aspirate or stool**.
3. **They respond well**: with correct weight-band dosing, cure rates are high, and **non-severe TB can now be treated in 4 months** (WHO 2022, SHINE trial, adopted in UCG 2023).

> **CLINICAL PEARL**
> The one thing never to forget: **every child with TB has an infectious adult somewhere, and every infectious adult has children at home.** Ask about the contact, screen every child in the household, treat the ones with disease, and give **TB preventive therapy** to the rest. This is the single most effective thing a clinician can do about childhood TB.

Term	Meaning
TB infection (latent TB infection, LTBI)	Infection with *M. tuberculosis* without symptoms or signs of disease; positive TST or IGRA; **not infectious**
TB disease	Symptoms, signs or radiological changes caused by *M. tuberculosis*
Presumptive TB	A child with symptoms or signs suggestive of TB who needs evaluation
Bacteriologically confirmed TB	*M. tuberculosis* detected (Xpert, culture, smear, LF-LAM)
Clinically diagnosed TB	Treated on clinical grounds (history, contact, examination, CXR, response) without bacteriological confirmation; most childhood TB
Pulmonary TB (PTB)	Disease of the lung parenchyma or tracheobronchial tree (UCG: if both pulmonary and extrapulmonary, classify as PTB)
Extrapulmonary TB (EPTB)	Any other site; isolated intrathoracic lymph node TB or pleural effusion without lung involvement is classed as EPTB (UCG 2023)
Drug-resistant TB	Rifampicin-resistant (RR), multidrug-resistant (MDR: R + H), pre-XDR (+ fluoroquinolone), XDR (+ bedaquiline or linezolid)

Term	Meaning
Contact	Anyone exposed to a person with TB; **household contact** = shared the same enclosed living space for ≥1 night or frequent or extended daytime periods in the 3 months before the index case started treatment

2. Epidemiology

- **Global**: in 2024 an estimated **10.7 million people** fell ill with TB, including **1.2 million children**, and **1.23 million died** (WHO fact sheet, 2026). TB is the world's leading cause of death from a single infectious agent. Children account for about 11% of cases but a larger share of deaths, mostly in under-fives who were **never diagnosed or treated**.
- **Uganda** is on WHO's list of high **TB/HIV** burden countries; about 4 in 10 adults with TB are HIV-positive in some districts Verify *district figure source*, and children living with HIV have a much higher risk of TB.
- **Ugandan practice**: TB is diagnosed and treated at HC III and above (DOTS sites); **Xpert MTB/RIF (Ultra)** is available at HC IVs and hospitals and through a **sample transport (hub) network**; stool Xpert (using simple one-step stool processing) is used for children; **TB/HIV is managed in one-stop clinics**; drug-resistant TB is managed at designated **MDR-TB treatment initiation centres** (e.g. Mulago, regional referral hospitals). **Digital adherence technologies** and **treatment supporters (DOT)** are standard (UCG 2023).

> **UGANDA CONTEXT**
> UCG 2023 adopted the **4-month regimen 2HRE(Z)/2RH for children 2 months to 16 years with non-severe disease** at **general hospital level and above**. At HC III and HC IV, children are usually started on the standard **2RHZE/4RH** regimen. If a child is clearly non-severe and stable, discuss with your senior whether they qualify for the shorter regimen.

3. Aetiology and Risk Factors

Cause: *Mycobacterium tuberculosis*, spread by **inhalation of droplet nuclei** from an adult or adolescent with pulmonary TB (cough, laughter, singing; poorly ventilated rooms). Young children rarely infect others.

Who gets TB disease after infection

Risk factor	Why it matters
Age <1 year (up to 30–40% progress) and **<2 years**	Immature cell-mediated immunity; highest risk of TBM and miliary TB. Risk falls in school-age children and rises again in adolescence
Household contact with pulmonary TB (especially smear- or Xpert-positive)	Strongest predictor of TB in a child
HIV infection	Blunted CD4 and IFN-γ response: more frequent, disseminated and atypical disease
Severe malnutrition	Same mechanism; TST often negative
Recent measles	Transient immune suppression can trigger TB
Diabetes, steroid use	Impaired immunity
No BCG	Loses protection against TBM and miliary TB

BCG (UNEPI: at birth) protects young children against **TB meningitis and miliary TB** (about 70–80% protection) but much less against pulmonary TB. Do not give BCG to a child with symptomatic HIV (UCG 2023).

4. Pathophysiology

Natural history after infection

1. **Inhalation** of droplet nuclei from an adult with pulmonary TB.
2. Bacilli multiply in alveolar macrophages and drain to the **hilar lymph nodes**: the **primary (Ghon) complex** = a small parenchymal focus + enlarged hilar nodes.
3. **Cell-mediated immunity** develops in 2–12 weeks (TST becomes positive). In most children, infection is contained (TB infection).
4. In young or immunocompromised children, containment fails:
    - **Lymph node disease**: enlarging hilar or paratracheal nodes compress bronchi (wheeze, collapse, hyperinflation) or erode into them (endobronchial TB, consolidation).
    - **Progressive primary disease**: caseation and cavitation (more in adolescents, adult-type disease).
    - **Haematogenous spread**: **miliary TB** and **TB meningitis** (within 2–6 months of infection, mainly <2 years), bones and joints (spine: 1–3 years after infection), kidneys.
    - **Pleural effusion** (school-age children; hypersensitivity reaction, 3–6 months after infection).
5. **Timeline (Wallgren)**: most disease appears **within 12 months** of infection; that is why TPT for recent contacts works.

Why children are paucibacillary and rarely infectious: most childhood disease is in lymph nodes and small lesions with few bacilli, and young children swallow rather than cough out sputum, so smears are negative and **young children rarely infect others** (adolescents with cavities can). Swallowed bacilli pass into the gut, which is why **gastric aspirate and stool** samples work.

Why HIV and malnutrition matter: TB control depends on **CD4 T cells and macrophage activation (IFN-γ)**. HIV and malnutrition blunt this, so disease is more frequent, disseminated and atypical, the **TST is often negative**, and mortality is higher. ART and nutritional rehabilitation are part of TB treatment.

Why we treat with several drugs for months

- Bacilli exist in different populations: rapidly dividing (killed by **isoniazid**), slowly or intermittently dividing (killed by **rifampicin**), and in acidic, inflamed sites (killed by **pyrazinamide**). **Ethambutol** protects against resistance when isoniazid resistance or high bacillary load is possible.
- The **intensive phase** (2 months, 3–4 drugs) kills most bacilli and prevents resistance; the **continuation phase** (2–10 months, H + R) sterilises the persisters.
- **Monotherapy** of active disease (e.g. isoniazid alone given as "IPT" to a child with undiagnosed TB) selects resistance: this is why we **exclude active TB before TPT**.

Why children need higher mg/kg doses than adults: children metabolise isoniazid and rifampicin faster; the 2010 WHO revision raised paediatric doses (H 10, R 15, Z 35, E 20 mg/kg) to reach adult-equivalent blood levels. Hepatotoxicity remains rare in children at these doses.

Inhaled droplet nuclei from infectious adult

bone

↓

Alveolar macrophages: Ghon focus

↓

Drain to hilar nodes: primary complex

↓

Cell-mediated immunity in 2-12 weeks

↓

Contained: TB infection, TST positive

Fails: young age, HIV, malnutrition, measles

↓

Node enlargement: bronchial compression or erosion

Progressive primary disease, cavities in adolescents

Haematogenous spread: miliary TB, TBM within 2-6 months

J

Reactivation later if immunity falls

Feature links

- Chronic cytokine release (TNF-α, IL-1) → persistent fever, night sweats, anorexia → weight loss, poor weight gain, faltering growth, lethargy, reduced play.
- Enlarged hilar or paratracheal nodes compressing a bronchus → monophonic wheeze unresponsive to bronchodilators, lobar collapse, hyperinflation; tracheal compression → stridor.
- Node eroding into a bronchus → endobronchial TB, consolidation, persistent cough.
- Caseation and cavitation in adolescents → productive cough, haemoptysis, smear-positive and infectious disease.
- Caseating peripheral nodes → firm, matted, non-tender neck nodes → fluctuant "cold abscess" → sinuses.
- Haematogenous seeding of the lungs → miliary pattern, hypoxaemia; of liver and spleen → hepatosplenomegaly; of the choroid → choroid tubercles.
- Rich focus rupturing into the subarachnoid space → basal exudate → cranial nerve palsies (VI, III, VII), obstructive hydrocephalus (bulging fontanelle, rising head circumference, sunsetting eyes), vasculitis → infarcts, hemiparesis, convulsions; CSF lymphocytes, high protein, low glucose.
- Hypersensitivity to TB antigen in the pleura → lymphocytic exudative effusion → stony dullness, reduced breath sounds.
- Vertebral body destruction → collapse → gibbus; paravertebral abscess or collapse → cord compression → leg weakness; psoas abscess.
- Pericardial involvement → effusion → muffled heart sounds, raised JVP, pulsus paradoxus, hepatomegaly; later constriction.
- Abdominal nodes and peritoneum → ascites, doughy abdomen, masses.
- Blunted cell-mediated immunity (HIV, SAM) → negative TST, atypical and disseminated disease.

5. Classification / Staging

Severity (WHO 2022)

Category	Includes	Treatment
Non-severe TB	**Peripheral lymph node TB**; **intrathoracic lymph node TB without airway obstruction**; **uncomplicated TB pleural effusion**; or **paucibacillary, non-cavitary disease confined to one lobe of the lungs without a miliary pattern**	4 months (2HRZ(E)/2HR)
Severe TB	Anything else: miliary TB, TB meningitis, osteoarticular TB, abdominal TB, pericardial TB, cavitary or extensive lung disease, airway compression, complicated effusion; also most clinicians treat **SAM or advanced HIV** with the standard 6-month regimen	6 months; TBM and bone or joint TB 12 months

By site: pulmonary vs extrapulmonary (definitions in section 1).

By drug susceptibility: drug-susceptible; RR, MDR, pre-XDR, XDR (section 1).

Treatment history categories (UCG 2023): **New** (no previous treatment or <1 month), **relapse**, **treatment after failure**, **treatment after loss to follow-up**, **history unknown**. In Uganda the **choice of regimen depends on rifampicin susceptibility (Xpert)**, not on treatment history.

TB meningitis stage (BMRC)

Stage	Criteria
I	GCS 15, no focal signs
II	GCS 11–14 or focal signs (e.g. cranial nerve palsy)
III	GCS ≤10

Treatment outcome definitions (UCG 2023)

Outcome	Meaning
Cured	Bacteriologically confirmed at start; smear or culture negative in the last month and on ≥1 previous occasion
Treatment completed	Completed without evidence of failure, without the negative results needed for "cured" (most children)
Treatment failed	Smear or culture positive at month 5 or later
Lost to follow-up	Did not start or interrupted for ≥2 consecutive months
Died / Not evaluated	–

6. Clinical Features

History

Each question has a reason.

- **TB contact**: anyone at home or a carer coughing, on TB treatment, or who died of a chest illness in the past 1–2 years? Smear- or Xpert-positive? Drug-resistant? → the strongest predictor of TB in a child; a DR contact changes the regimen and the TPT choice.
- **Symptoms** (any for ≥2 weeks): **cough** not improving, **fever**, **night sweats** → presumptive TB.
- **Weight loss or poor weight gain** (check the Child Health Card growth chart) → presumptive TB; faltering growth is a TB symptom.
- **Reduced playfulness, lethargy, poor feeding** → TB in young children.

- **Site-specific**: neck swellings (lymph node TB); headache, vomiting, irritability, convulsions, drowsiness (TBM → LP); back hump or refusal to walk (spinal TB); abdominal swelling (abdominal TB); joint swelling; breathlessness and leg swelling (pericardial TB).
- **HIV status** of child and mother; on ART? which regimen? cotrimoxazole? → TB/HIV management, drug interactions, LF-LAM.
- **Previous TB treatment or TPT**, adherence and outcome → relapse, resistance.
- **Immunisation**: BCG (scar at right upper arm), measles (recent measles can trigger TB).
- **Nutrition**: weight trend, oedema, feeding → SAM management; slower response.
- **Response to previous antibiotics or antimalarials** → pneumonia or fever not responding is a TB clue.

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: observe first, listen to the chest while the child is quiet, and leave distressing parts (fundoscopy, ears, throat) to last.

Introduction, consent and preparation

- Introduce yourself; confirm identity; explain; consent from the parent and assent from the child; **hand hygiene**.
- If an adolescent or adult carer is coughing: give a mask and examine in a ventilated space.
- **Position and exposure**: on the parent's lap; expose the chest, abdomen and back (spine) in a warm room.

General inspection (end of the bed)

Sign	What it means
Lethargic, not playing	Systemic TB; TBM if drowsy or irritable
Fast breathing, indrawing	Miliary TB, effusion, extensive disease
Wasting, oedema	Malnutrition: TB evaluation and SAM care
Abnormal posture, head size, sunsetting eyes	TBM with hydrocephalus
Gibbus visible on the back	Spinal TB

Vital signs and growth: temperature, RR, HR, **SpO_2**; **weight, height, Z-scores (WHZ, WAZ, HAZ), MUAC, oedema**, plotted on the growth chart; head circumference in infants.

Hands, face and neck

- **Hands**: pallor, clubbing (chronic lung disease, HIV).
- **Face and mouth**: oral thrush and parotid swelling (HIV).
- **Lymph nodes**: cervical (and axillary): **firm, matted, non-tender, may fluctuate as a "cold abscess" or form sinuses**; generalised lymphadenopathy (HIV).
- **BCG scar** on the right upper arm.

Respiratory system

- **Inspection**: RR, indrawing, asymmetry.
- **Palpation**: trachea, expansion.
- **Percussion**: **stony dullness** (effusion); dullness (consolidation, collapse).
- **Auscultation**: crackles, bronchial breathing, **monophonic wheeze** (node compression), reduced breath sounds.

Completing the examination

- **Abdomen**: hepatosplenomegaly (miliary), ascites, doughy abdomen, masses (abdominal TB).

- **Spine and joints**: **gibbus** (angular deformity), tenderness, swollen joints, sinuses.
- **Heart**: muffled heart sounds, raised JVP, pulsus paradoxus, hepatomegaly (pericardial effusion).
- **Neurology**: consciousness (GCS or AVPU), neck stiffness, bulging fontanelle, **cranial nerve palsies (VI, III, VII)**, hemiparesis.
- **Fundoscopy, last**: papilloedema, choroid tubercles (miliary TB).
- **Thank**, **wash hands**, **summarise**.

Document these positives	And these negatives
Weight, Z-scores, growth trend	"No neck stiffness, no cranial nerve palsy, GCS 15"
Lymph nodes (site, size, matted, fluctuant, sinus)	"No gibbus; no joint swelling"
Chest signs; SpO_2	"No hepatosplenomegaly or ascites"
BCG scar present or absent	"Heart sounds normal; no raised JVP"

Red flags

RED FLAG

TB meningitis until proven otherwise: a child (often <5 y) with **fever and irritability or lethargy for more than a week**, vomiting, headache, **cranial nerve palsy**, convulsions, reduced consciousness, especially with a TB contact. Do an LP, start treatment (anti-TB + steroids) urgently: outcome depends on the stage at which treatment starts.

RED FLAG

Other emergencies: **airway compression** by nodes (stridor, severe wheeze, lobar collapse); **miliary TB** with hypoxaemia; **pericardial tamponade**; **spinal TB with leg weakness** (cord compression); **severe malnutrition** with TB; **drug-induced hepatitis** (jaundice, vomiting, abdominal pain on treatment).

Thinking like a clinician

- **"Persistent, unremitting, not improving"** is the TB pattern: symptoms for >2 weeks that do not settle with antibiotics or antimalarials.
- **Contact + symptoms = treat** in a young child, even with a negative Xpert, if the picture fits (or use the treatment decision algorithm).
- **Faltering growth is a TB symptom.** A child in the nutrition unit who is not gaining weight on RUTF needs TB evaluation.
- **TB and HIV travel together.** Test every child with TB for HIV, and screen every child with HIV for TB at every visit.
- **Wheeze that does not respond to bronchodilators in an infant with a TB contact** = node compression until proven otherwise.
- **A negative TST does not exclude TB** (malnutrition, HIV, miliary TB, TBM, early infection); a positive one shows infection, not disease.
- **Lymphocytic exudate pleural effusion in a school-age child** = TB until proven otherwise.
- **Adolescents can have adult-type cavitary disease** and are infectious: isolate and test sputum.
- **Before starting TPT, always exclude active disease** with a symptom screen; giving TPT to a child with TB creates resistance.
- **Common student mistakes**: waiting for sputum from a toddler; keeping the same dose for 6 months as the child gains weight; forgetting to screen the siblings; giving 4 months to TBM.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Recurrent or unresolved bacterial pneumonia	Acute onset, fever, responds to antibiotics	CXR, response to treatment
HIV-related lung disease (LIP, bronchiectasis, PCP)	HIV-positive, clubbing, parotid swelling (LIP), severe hypoxaemia (PCP)	HIV test, CXR pattern
Asthma	Recurrent wheeze, responds to bronchodilator, atopy	Bronchodilator trial
Foreign body	Sudden onset, unilateral signs	CXR, bronchoscopy
Pertussis	Paroxysms, whoop, apnoea	Clinical
Malnutrition from food insecurity	Improves with feeding, no contact or fever	Response to nutritional rehabilitation
Malignancy (lymphoma, Kaposi sarcoma in HIV, leukaemia)	Rapid node growth, hard nodes, cytopenias	FNA or biopsy, FBC
Lymphadenitis (bacterial, BCG adenitis, non-tuberculous mycobacteria)	Acute tender node; BCG adenitis in the axilla on the side of the BCG (right axilla after right-arm BCG)	FNA
Bacterial or cryptococcal meningitis (vs TBM)	Acute onset (bacterial); HIV with headache (cryptococcal)	CSF Gram stain, CrAg
Heart failure, rheumatic carditis (vs pericarditis)	Murmur, rheumatic history	Echo

8. Clinical Workup

Confirm when you can, treat when you must. "Available at" is the lowest level that usually offers the test

Verify *compiled from the source notes, not an MoH list.*

Bedside

Test	Why / how	Expected finding	How it changes management	Available at
Weight and growth chart	Every visit	Weight loss or flat curve = TB symptom	Dose by weight band; response to treatment	All levels
SpO_2	Pulse oximeter	Hypoxaemia: miliary TB, severe disease	Oxygen; admission	HC IV and above
HIV rapid test (DNA PCR <18 months)	All children with presumptive TB	Positive	LF-LAM, ART, cotrimoxazole	Rapid test all levels; PCR via hub
Tuberculin skin test (TST)	0.1 mL (5 TU) PPD intradermally on the forearm; read at **48–72 h** (transverse induration)	**≥10 mm** positive; **≥5 mm** in HIV-positive or severely malnourished children	Shows infection, not disease; supports a clinical diagnosis; **not required** before TPT in <5 y or HIV	Variable; PPD supply often limited Verify

Test	Why / how	Expected finding	How it changes management	Available at
Urine LF-LAM	Urine strip, 25 min; HIV-positive children with TB signs or symptoms who are seriously ill or have advanced HIV, or inpatients	Positive	Treat TB	HIV clinics
Sample collection	Expectorated (older), **induced sputum** (hypertonic saline nebulisation + suction), **gastric aspirate** (early morning, fasting, NG tube), **nasopharyngeal aspirate**	–	Samples for Xpert and culture	HC IV and hospitals
Stool for Xpert	Fresh stool in a container; easy sample in young children (WHO 2022)	–	Xpert	Collected at any level; tested via hub

How to collect a gastric aspirate: fasting overnight (or ≥4 h in infants), early morning before the child gets up, NG tube, aspirate 5–10 mL (or instil 10 mL sterile water and aspirate), send promptly (neutralise if delayed).

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Xpert MTB/RIF Ultra	Every child with presumptive TB: sputum (expectorated or induced), **gastric aspirate, NPA, stool**; CSF, node aspirate, pleural fluid for EPTB	MTB detected (including "trace", which counts as positive in children); **rifampicin resistance** detected or not	Positive → treat; **RR** → refer to MDR-TB centre. **Negative does not exclude TB** (Ultra sensitivity in children about 60–70% against culture)	HC IV and hospitals, or via hub transport
LFTs	Baseline if liver disease, HIV, or jaundice on treatment	ALT >3× ULN with symptoms or >5× without	Stop hepatotoxic drugs; reintroduce sequentially	General hospital and above Verify
CD4, viral load	HIV-positive	Advanced HIV	ART timing, prophylaxis	Via hub

Imaging

Test	When	Expected finding	How it changes management	Available at
Chest X-ray (AP + lateral if possible)	Presumptive PTB, contacts with symptoms, HIV	**Hilar or paratracheal lymphadenopathy**, consolidation, collapse, **miliary pattern**, cavities (adolescents), **pleural effusion**	Supports diagnosis; classifies severe vs non-severe	HC IV (some) and hospitals
Abdominal ultrasound	Abdominal TB, HIV	Enlarged abdominal nodes, ascites, splenic micro-abscesses	Supports EPTB	Hospitals
Spine X-ray / MRI	Back pain, gibbus	Vertebral collapse, disc space loss, paravertebral abscess	Spinal TB (12 months); neurosurgery if cord compression	X-ray at hospitals; MRI at NRH and private
Echocardiography	Suspected pericardial TB	Pericardial effusion	Pericardiocentesis; steroids	RRH, NRH
CT head	TBM	Hydrocephalus, basal enhancement, infarcts	VP shunt	NRH, CURE Mbale, private

Specialised / referral-level

Test	When	Expected finding	How it changes management	Available at
Lumbar puncture	Suspected TBM	**Clear CSF, lymphocytes 50–500/µL, protein high (>1 g/L), glucose low** (CSF:blood <0.5), Xpert Ultra on CSF; CrAg if HIV	TBM regimen + steroids	Hospitals
Fine-needle aspirate of a node	Lymphadenopathy >2 cm, persistent	Caseation, AFB, Xpert positive	Diagnosis of node TB	Hospitals
Pleural tap	Effusion	Straw-coloured lymphocytic exudate, protein >30 g/L; Xpert (low yield)	TB effusion vs empyema	Hospitals
IGRA	Contact without symptoms; supportive in diagnosis	Positive = infection	Supports decision to treat	Private and research laboratories Verify
Culture and DST (liquid culture, line probe assay, Xpert XDR)	RR-TB, failure, contacts of DR-TB	Resistance pattern	Second-line regimen	National reference laboratory (CPHL/NTRL)

Putting it together

Model one-line summary

> "A 3-year-old boy weighing 11 kg (WAZ −2.4, WHZ −1.8, HAZ −1.5), HIV-negative, BCG scar present, whose grandmother started treatment for Xpert-positive pulmonary TB a month ago, with 5 weeks of cough, intermittent fever, night sweats and weight loss despite amoxicillin, with right hilar lymphadenopathy on CXR and stool Xpert Ultra 'MTB detected, rifampicin resistance not detected': **bacteriologically confirmed pulmonary TB (intrathoracic lymph node disease without airway obstruction)**."

Problem list

1. Pulmonary TB, drug-susceptible, likely non-severe.
2. Moderate underweight (WAZ −2.4) and borderline wasting.
3. Household TB exposure: other children need screening and TPT.

Working diagnosis: *drug-susceptible, bacteriologically confirmed, non-severe pulmonary TB in an HIV-negative child with undernutrition.*

Model case presentation (ward round)

> "Kato is a 3-year-old boy from Kisenyi weighing 11 kg (WAZ −2.4, WHZ −1.8, HAZ −1.5; MUAC 12.8 cm, no oedema), fully immunised per UNEPI with a BCG scar, HIV-negative (rapid test today), on family diet. He has had a cough for 5 weeks, intermittent fevers and night sweats, and has lost 1 kg on his growth chart despite a course of amoxicillin. His grandmother, who sleeps in the same room, started treatment for Xpert-positive pulmonary TB a month ago; two siblings aged 18 months and 6 years live with them. On examination: T 37.9 °C, RR 32, SpO_2 96%, small right cervical nodes, few crackles at the right upper zone, no hepatosplenomegaly, no neck stiffness, no gibbus. CXR: right hilar lymphadenopathy with a small right upper-lobe opacity, no cavity, no airway compression. Stool Xpert Ultra: MTB detected, rifampicin resistance not detected. Assessment: **bacteriologically confirmed, drug-susceptible, non-severe pulmonary TB**

> **with moderate underweight**. Plan: register and start **2HRZE/2HR** (4-month regimen) with RHZ 75/50/150 × 2 + E 100 mg × 2 daily, then RH × 2, pyridoxine; treatment supporter (mother); nutritional support; screen both siblings and give **3HR** to the 18-month-old and **3HP** to the 6-year-old if symptom-free; monthly weight and band review."

EXAM FAVOURITE

Worked example 1: dosing by weight band. An 11 kg child is in the **8–11.9 kg band → 2 tablets of RHZ 75/50/150** (= R 150 mg, H 100 mg, Z 300 mg, i.e. R 13.6, H 9.1, Z 27 mg/kg) plus **ethambutol 100 mg × 2** (18 mg/kg) daily in the intensive phase, then **RH 75/50 × 2** daily. Dispersible: dissolve in a small amount of water and give at once. If the child reaches **12 kg** next month, move to **3 tablets**. (The pyrazinamide is just under 30 mg/kg at the top of the band; that is accepted within the band design.)

EXAM FAVOURITE

Worked example 2: reading a CSF. A 2-year-old with 10 days of fever, irritability, vomiting and a left VI nerve palsy. CSF: clear, WBC 220/µL (85% lymphocytes), protein 2.1 g/L, glucose 1.2 mmol/L (blood 5.0; ratio 0.24), Gram stain negative, Xpert Ultra on CSF "trace". **Interpretation**: lymphocytic meningitis with very high protein and very low glucose plus a cranial nerve palsy and subacute course = **TB meningitis, BMRC stage II**. Start **2RHZE/10RH** (or 6HRZ-Eto) plus **prednisolone 2 mg/kg/day**, check for hydrocephalus (CT if available), test for HIV, and screen the household.

Treatment decision scores (when bacteriology is negative or unavailable)

WHO 2022 treatment decision algorithms (TDAs) for children <10 years with presumptive pulmonary TB, used at facilities with or without CXR: the clinician scores symptoms (cough, fever, lethargy, weight loss, haemoptysis, night sweats), signs (lymph nodes, tachycardia, tachypnoea) and, where available, CXR features; **a total score >10 means treat for TB**. They are recommended for young children who cannot be confirmed, and they increase diagnosis without many false positives.

Keith Edwards score (older, still taught in Ugandan and East African units): a total of **≥7 suggests TB and justifies treatment**.

Feature	0	1	2	3	4
Duration of illness	<2 weeks	2–4 weeks		>4 weeks	
Nutrition (weight-for-age)	>80%	60–80%		<60%	
Family history of TB	None		Reported by family	Proven (sputum-positive)	
Positive tuberculin test				Yes	
Malnutrition not improving after 4 weeks of treatment				Yes	
Unexplained fever and night sweats not responding to malaria treatment			Yes		
Pneumonia or chest disease not responding to antibiotics			Yes		
Lymph nodes (neck, axilla), abdominal mass or ascites				Yes	
CNS signs with abnormal CSF				Yes	
Angle deformity of the spine					Yes

Feature	0	1	2	3	4
Joint or bone swelling or sinuses				Yes	

9. Management

Immediate / emergency

Most children with TB are not acutely unstable, but some are:

1. **ETAT triage**: respiratory distress (miliary TB, effusion, airway compression), shock (sepsis, tamponade), **coma or convulsions (TBM)**, severe malnutrition.
2. **Oxygen** for SpO_2 <90%; position; check glucose.
3. **Convulsions**: diazepam 0.5 mg/kg rectally or 0.2–0.3 mg/kg IV.
4. **Altered consciousness with fever**: treat as meningitis (ceftriaxone ± antimalarial) while you plan an LP; if the picture is subacute, add anti-TB treatment and steroids early.
5. **Infection control**: if an adolescent or adult carer is coughing, give masks, ventilate, separate (cough triage).
6. **Weigh** the child: every TB drug dose depends on the weight band.

ABCDE: **A** stridor from node compression → senior, steroids; **B** oxygen, effusion → tap; **C** muffled heart sounds and raised JVP → tamponade → pericardiocentesis; **D** GCS, glucose, convulsions, signs of raised ICP or hydrocephalus; **E** temperature, nodes, spine, weight.

Definitive

Principles

1. **Confirm when you can, treat when you must**: collect the best sample (sputum, gastric aspirate, stool) for Xpert, but **do not delay treatment** in a sick child with a convincing picture.
2. **Classify**: site (PTB, EPTB), severity (non-severe, severe), drug susceptibility (Xpert), HIV status, nutrition.
3. **Choose the regimen** and **dose by weight band** using dispersible paediatric FDCs.
4. **Directly observed or supported treatment** (treatment supporter, digital adherence technologies), with a **TB treatment card** and register entry at the DOTS clinic.
5. **Treat HIV and malnutrition** alongside TB.
6. **Trace the source and screen all contacts**; give **TPT** to eligible contacts.
7. **Notify** (the TB register is the notification).

Paediatric dispersible FDC weight bands (WHO; Uganda NTLP)

Weight band	Intensive phase: RHZ 75/50/150	+ Ethambutol 100 mg	Continuation: RH 75/50
4–7.9 kg	**1** tablet	1 tablet	**1** tablet
8–11.9 kg	**2**	2	**2**
12–15.9 kg	**3**	3	**3**
16–24.9 kg	**4**	4	**4**
≥25 kg	Use **adult FDCs** (RHZE 150/75/400/275; RH 150/75) by adult weight bands		

Children **<4 kg** (young infants): mg/kg dosing with specialist input. **Re-weigh monthly** and move bands as weight rises.

Regimens for drug-susceptible TB

Type of TB	Regimen	Duration	Where (Uganda)
Non-severe TB, children **3 months–16 years** (WHO 2022; UCG 2023: 2 months–16 years)	**2HRZ(E) / 2HR** (E added in high-HIV or high isoniazid-resistance settings, i.e. Uganda)	**4 months**	General hospital and above (UCG)
All other forms of PTB and EPTB (except TBM, bone or joint)	**2RHZE / 4RH**	**6 months**	HC III and above
TB meningitis and **osteoarticular (bone or joint) TB**	**2RHZE / 10RH**	**12 months**	HC IV and above
TBM alternative (WHO 2022, children and adolescents with bacteriologically confirmed or clinically diagnosed DS-TBM)	**6HRZ-Eto** (high-dose H and R, Z, ethionamide daily)	**6 months**	Specialist centres; not yet standard in UCG 2023
Adolescents ≥12 y, >40 kg (WHO; UCG for adults)	2HPMZ/2HPM (rifapentine-moxifloxacin)	4 months	Specialist; HIV-positive only if CD4 >100

UGANDA CONTEXT

Which children qualify for 4 months? Only clearly **non-severe** disease (section 5) with no TBM, miliary, osteoarticular, abdominal, pericardial or extensive or cavitary disease, and no airway compression. In practice many Ugandan clinicians **also give 6 months** to children with **severe acute malnutrition** or **advanced HIV**, because these groups were few in the SHINE trial. Check with your senior and the NTLP chart. Note the age range: WHO 2022 says 3 months–16 years; UCG 2023 says 2 months–16 years.

Adjunctive treatment

- **Corticosteroids**: **TB meningitis** (prednisolone **2 mg/kg/day, max 60 mg**, for **4 weeks**, then taper over 2 weeks; UCG 2023: 1–2 mg/kg, max 60 mg; dexamethasone IV if unable to swallow), **TB pericarditis**, and **airway compression** by nodes.
- **Pyridoxine** with isoniazid.
- **Nutritional support**: RUTF, F-75 or F-100 per the WHO 2023 wasting/SAM guideline if malnourished; food support.
- **Analgesia, antipyretics**, physiotherapy (spinal TB).
- **Surgery**: drainage of cold abscesses, spinal decompression for cord compression, ventriculoperitoneal shunt for hydrocephalus in TBM (NRH, CURE Mbale).

TB meningitis: step by step

1. **Suspect early** (subacute fever, irritability, vomiting, headache, cranial nerve palsy, convulsions, reduced consciousness).
2. **LP** (if no signs of raised ICP or focal deficit that make it unsafe; otherwise treat empirically): cell count, protein, glucose (with blood glucose), Gram stain, **Xpert Ultra**, CrAg (if HIV).
3. **Stage** (BMRC, section 5).
4. **Start anti-TB treatment immediately**: 2RHZE/10RH (or 6HRZ-Eto) + **prednisolone 2 mg/kg/day** (max 60 mg) × 4 weeks then taper over 2 weeks.
5. **Supportive**: airway, nutrition (NG), seizure control, head positioning, monitor for **hydrocephalus** (bulging fontanelle, increasing head circumference, sunsetting eyes), hyponatraemia.
6. **HIV-positive**: start ART **4–8 weeks** after TB treatment (to reduce IRIS-related deaths).

TB and HIV co-infection

Issue	What to do (Uganda consolidated HIV guidelines; UCG 2023; WHO)
HIV testing	Every child with presumptive or confirmed TB
Timing of ART	Start **TB treatment first**, then **ART within 2 weeks** (as soon as tolerated); **TBM: delay ART 4–8 weeks**
Already on ART	Continue ART; adjust for rifampicin interactions
Dolutegravir (DTG)	With rifampicin, give **DTG twice daily** (both pDTG 10 mg dispersible and DTG 50 mg), continue twice daily for **2 weeks after rifampicin stops**
Lopinavir/ritonavir	**Super-boost** with extra ritonavir (LPV:RTV 1:1) or switch to a DTG-based regimen; or use rifabutin (specialist)
Efavirenz (older regimens in UCG tables)	Can continue at standard dose with rifampicin (≥3 y)
Nevirapine	Avoid with rifampicin (switch)
Cotrimoxazole prophylaxis	All HIV/TB co-infected children
IRIS	Worsening fever, nodes or chest signs after starting ART: continue both; prednisolone for moderate–severe paradoxical IRIS (specialist)

> **UGANDA CONTEXT**
> The **TB/HIV ART table in UCG 2023** still shows **EFV- and NVP-based** regimens for children and triple-NRTI options for under-3s; Uganda's current **consolidated HIV guidelines** use **DTG-based ART** for children ≥4 weeks and ≥3 kg (pDTG 10 mg dispersible; DTG 50 mg ≥20 kg). On today's wards, the practical rule is: **child on DTG + rifampicin → DTG twice daily**. Check the child's ART regimen against the current HIV guideline, not only the UCG table.

Drug-resistant TB (overview)

- Suspect if: contact of DR-TB, Xpert rifampicin resistance, failure on first-line, relapse.
- **Refer to an MDR-TB treatment initiation centre**; culture and DST.
- WHO 2022 recommends **all-oral regimens** for children of all ages, including **bedaquiline and delamanid**; the 6-month **BPaLM** regimen is for **≥14 years**; children receive 9-month or longer individualised regimens.

Contact screening and TB preventive therapy (TPT)

Who to screen: every household and close contact of a person with TB, especially **children <5 years** and **HIV-positive** people.

Algorithm (UCG 2023, WHO 2024):

1. **Symptom screen** the child: **cough, fever, poor weight gain** (and lethargy, night sweats).
2. **Any symptom** → evaluate for TB disease (CXR, Xpert, clinical assessment).
3. **No symptoms** → TB disease unlikely → check contraindications (active hepatitis, alcohol use in adolescents, peripheral neuropathy) → **give TPT**. TST or IGRA is **not required** before TPT in children <5 years or people with HIV.
4. **HIV-positive children <1 year** receive TPT **only if they have a TB contact** (UCG 2023); HIV-positive children ≥1 year receive TPT as part of HIV care once TB is excluded.

TPT regimens

Regimen	Who	Dosing
3HP (isoniazid + rifapentine **once weekly for 12 weeks**)	**≥2 years** (Uganda's preferred option for most)	Child FDC **HP 300/300**: **10–15.9 kg 1 tab; 16–23.9 kg 1.5; 24–30.9 kg 2; 31–34.9 kg 2.5; ≥35 kg 3** (UCG chart); or single drugs H 100 mg + P 150 mg tablets per NTLP chart. Pyridoxine weekly or daily
3HR (isoniazid + rifampicin **daily for 3 months**)	**Any age** (good for <2 y)	**RH 75/50 dispersible**: 4–7.9 kg 1 tab; 8–11.9 kg 2; 12–15.9 kg 3; 16–24.9 kg 4; ≥25 kg adult RH
6H (isoniazid **daily for 6 months**)	Any age; HIV-positive (also pregnant women)	**Isoniazid 10 mg/kg (7–15), max 300 mg** + pyridoxine
1HP (daily isoniazid + rifapentine for 28 days)	**≥13 years**	H 300 mg + P 600 mg daily
6Lfx (levofloxacin daily for 6 months)	Contacts of **MDR/RR-TB** (WHO 2024)	≈15–20 mg/kg/day by weight band (specialist)

UGANDA CONTEXT

UCG 2023 says "do not use TPT in contacts of MDR-TB"; **WHO 2024** now recommends **6 months of levofloxacin** for household contacts of MDR/RR-TB after excluding disease. In Uganda, such children should be **referred to (or discussed with) the MDR-TB centre**, screened and followed up for 2 years, and given levofloxacin where the programme provides it.

Interactions: rifapentine and rifampicin-based TPT interact with ART (PIs, NVP); **3HP can be given with DTG** at the standard once-daily dose (WHO); with LPV/r use 6H.

TPT follow-up: monthly (or at each dose pick-up) for adherence, side effects and new TB symptoms; record completion.

Drug table

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Isoniazid (H)	**10 mg/kg (7–15)**, max 300 mg	Oral (in FDC)	Once daily	Whole regimen; 6 months as 6H TPT	Hepatitis, **peripheral neuropathy** (give **pyridoxine**), rash
Rifampicin (R)	**15 mg/kg (10–20)**, max 600 mg	Oral (in FDC)	Once daily	Whole regimen; 3 months as 3HR	**Orange urine and tears**, hepatitis, flu-like symptoms, thrombocytopenia; **enzyme inducer** (ART, contraceptives, anticonvulsants, warfarin)
Pyrazinamide (Z)	**35 mg/kg (30–40)**, max 2,000 mg	Oral (in FDC)	Once daily	Intensive phase (2 months; 6 months in 6HRZ-Eto)	Hepatitis, arthralgia, raised urate
Ethambutol (E)	**20 mg/kg (15–25)**, max 1,200 mg	Oral (100 mg dispersible)	Once daily	Intensive phase	**Optic neuritis** (rare in children at this dose; safe in young children who cannot report vision)

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Pyridoxine (vitamin B6)	UCG 2023: **25 mg daily** with isoniazid (many units give 12.5 mg in infants)	Oral	Once daily	With isoniazid	Prevents neuropathy, especially in HIV, malnutrition, breastfed infants
Ethionamide (Eto) (TBM 6-month regimen)	17.5–22.5 mg/kg, max 750 mg	Oral	Once daily	6 months	GI upset, hypothyroidism
Rifapentine (P) (3HP, 1HP)	By weight band (3HP FDC HP 300/300 table above); 1HP: P 600 mg daily	Oral	Weekly (3HP); daily (1HP)	12 weeks (3HP); 28 days (1HP)	≥2 y (3HP), ≥13 y (1HP); interacts with PIs and NVP; fine with DTG
Levofloxacin (MDR-TB contacts)	≈15–20 mg/kg/day by weight band	Oral	Once daily	6 months	Specialist; not in UCG 2023
Prednisolone (TBM, pericarditis, airway compression, IRIS)	**2 mg/kg/day, max 60 mg** (UCG 2023: 1–2 mg/kg)	Oral	Once daily	**4 weeks, then taper over 2 weeks**	Dexamethasone IV if unable to swallow
Diazepam (convulsions)	0.5 mg/kg rectally or 0.2–0.3 mg/kg IV	PR/IV	Once, repeat if needed Verify	–	Respiratory depression
Cotrimoxazole prophylaxis (TB/HIV)	Per HIV guideline Verify *weight-band dose*	Oral	Once daily	Ongoing	Rash

Adult doses: adult FDCs (RHZE 150/75/400/275; RH 150/75) by adult weight bands from 25 kg Verify *adult band table against UCG 2023*.

Referral criteria

Level	What to do	Refer up when
VHT / community	Identify coughing adults and symptomatic child contacts; adherence support	Any child with TB symptoms or a contact → HC III
HC III (DOTS site)	Symptom screen, collect sputum or stool for hub transport, HIV test, start **2RHZE/4RH**, TPT for contacts, monthly weights	Severe TB, TBM suspicion, sick or malnourished child, diagnostic doubt, RR on Xpert → HC IV or hospital
HC IV	Xpert, CXR at some, gastric aspirate; 6- and 12-month regimens	TBM, miliary or complicated disease, airway compression, hepatotoxicity, non-response, MDR → hospital or RRH
General Hospital	**4-month regimen** for non-severe TB (UCG), LP, inpatient care	Surgery, pericardiocentesis, hydrocephalus, MDR → RRH or MDR centre
RRH / MDR-TB treatment initiation centre	DR-TB treatment, culture and DST via CPHL/NTRL, echocardiography	Neurosurgery (VP shunt), spinal decompression → NRH or CURE Mbale
NRH (Mulago)	Neurosurgery, MRI, PICU	n/a

Level split derived from UCG 2023 and the source notes Verify.

Monitoring and follow-up

When	What
Every visit (at least monthly)	**Weight (adjust band)**, symptoms, adherence (treatment card, supporter), side effects (jaundice, vomiting, rash, numbness, vision), growth
End of month 2 (bacteriologically confirmed PTB who can give sputum)	Smear or Xpert: positive → Xpert for rifampicin resistance, continue continuation phase, review adherence (UCG)
Month 5 and 6	Smear: positive at month 5 or later = **treatment failure** → Xpert or DST, refer
Hepatotoxicity	Jaundice, vomiting, abdominal pain, hepatomegaly → **stop H, R, Z**, check LFTs; reintroduce one by one once LFTs are normal (specialist)
Interruptions	Trace; manage per NTLP (resume or restart depending on duration and phase)
Outcome	Record cured, completed, failed, LTFU, died or not evaluated for every child

Expect weight gain and fewer symptoms within 1–2 months. **Not improving**: check dose (band), adherence, HIV, malnutrition, wrong diagnosis, **drug resistance** (source case DR-TB? Xpert), IRIS.

10. Complications

Disease-related

Complication	Notes
TB meningitis sequelae	Hydrocephalus, blindness, deafness, cerebral palsy, epilepsy, intellectual disability, death
Airway compression	Lobar collapse, bronchiectasis, obstructive emphysema
Miliary TB	Respiratory failure, meningitis, death
Spinal TB (Pott's)	Gibbus, paraplegia (cord compression), cold abscess (psoas)
Pericardial TB	Tamponade, constrictive pericarditis
Post-TB lung disease	Bronchiectasis, fibrosis, chronic cough, reduced lung function (UCG 2023 post-TB management)
Growth faltering, school loss, family poverty	Social support, school letters

Treatment-related

Complication	Notes
Drug toxicity	Hepatitis (stop drugs, LFTs), neuropathy (pyridoxine), optic neuritis (ethambutol), rash, SJS
IRIS in HIV	Paradoxical worsening after ART
Drug resistance	Poor adherence, under-dosing, monotherapy (TPT given to a child with undiagnosed TB)
Drug interactions	Rifampicin lowers DTG, LPV/r, NVP and other drug levels

11. Prognosis

- With correct weight-band dosing, **cure rates are high**; expect weight gain and fewer symptoms within 1–2 months.

- **Non-severe TB**: the SHINE trial (NEJM 2022; 1,204 children in Uganda, Zambia, South Africa and India) found 4 months non-inferior to 6 months (unfavourable outcome 3% in both arms).
- **TB meningitis**: prognosis depends on the stage at the start of treatment: **stage I usually recovers fully; stage III often leaves disability or dies**.
- **Deaths** are mostly in under-fives who were **never diagnosed or treated**.
- HIV and malnutrition raise mortality.

12. Prevention

- **BCG at birth** (UNEPI; not for symptomatic HIV-positive infants).
- **Find and treat adults** with TB early (cough triage, Xpert).
- **TPT for contacts and people with HIV** (section 9).
- **Infection control**: ventilation, cough etiquette, masks for coughing adults and adolescents, separating infectious adolescents on the ward.
- **ART and nutrition** reduce TB risk in HIV and malnutrition.
- **Neonate born to a mother with infectious TB**: if the baby is well, give **TPT (isoniazid 6 months or 3HR)**, then BCG after TPT (or per national guidance); continue breastfeeding; mother treated and wears a mask until non-infectious.

13. Special Populations

- **Infants and children <2 years**: highest risk of TBM and miliary TB; use 3HR or 6H for TPT (3HP only from 2 years); **<4 kg**: mg/kg dosing with specialist input.
- **Neonate of a mother with infectious TB**: TPT, then BCG; continue breastfeeding.
- **HIV-positive children**: TB first then ART within 2 weeks (TBM 4–8 weeks); DTG twice daily with rifampicin; cotrimoxazole; LF-LAM; TST ≥5 mm; TPT under 1 year only with a contact; IRIS.
- **Severe acute malnutrition**: TST ≥5 mm; many clinicians give 6 months rather than 4; nutritional rehabilitation; slower response.
- **Adolescents**: adult-type cavitary disease and infectious; isolate; ≥25 kg adult FDCs; 1HP from 13 years; 2HPMZ/2HPM if ≥12 y and >40 kg; BPaLM for DR-TB from 14 years; ask about alcohol before TPT.
- **Contacts of MDR-TB**: UCG 2023 no TPT vs WHO 2024 levofloxacin; refer to the MDR-TB centre; follow for 2 years.

14. Key Points

High-yield points

- Every child with TB has an infectious adult somewhere: trace the source, screen the household.
- Suspect TB: cough or fever >2 weeks, poor weight gain, lethargy, contact.
- Samples: induced sputum, gastric aspirate, NPA or **stool** for Xpert Ultra; **negative Xpert does not exclude TB**.
- Doses **10-15-35-20**; dispersible FDCs by weight band; re-weigh monthly.
- 4 months for non-severe, 6 months standard, 12 months for TBM and bone TB.
- TBM: prednisolone 2 mg/kg/day (max 60 mg) for 4 weeks, then taper.
- TB/HIV: ART within 2 weeks (TBM 4–8 weeks); DTG twice daily with rifampicin.
- TPT only after excluding active disease: 3HP (≥2 y), 3HR (any age), 6H, 1HP (≥13 y).
- Faltering growth is a TB symptom.

- Test every child with TB for HIV, and screen every child with HIV for TB.

Numbers & doses

Item	Value
Suspect TB	Cough or fever **>2 weeks**, poor weight gain **1 month**, contact
Isoniazid	**10 mg/kg** (7–15), max 300 mg
Rifampicin	**15 mg/kg** (10–20), max 600 mg
Pyrazinamide	**35 mg/kg** (30–40)
Ethambutol	**20 mg/kg** (15–25)
Pyridoxine	**25 mg daily** (UCG)
FDC bands (RHZ 75/50/150; RH 75/50)	**4–7.9 kg: 1; 8–11.9: 2; 12–15.9: 3; 16–24.9: 4; ≥25: adult**
Non-severe TB	**2HRZ(E)/2HR = 4 months** (3 months–16 years)
Standard	**2RHZE/4RH = 6 months**
TBM and bone or joint TB	**2RHZE/10RH = 12 months** (WHO alternative for TBM: 6HRZ-Eto)
TBM steroid	Prednisolone **2 mg/kg/day (max 60 mg) × 4 weeks + 2-week taper**
TST positive	**≥10 mm**; **≥5 mm** HIV or SAM
WHO TDA	Score **>10** → treat
Keith Edwards	Score **≥7** → treat
ART timing	Within **2 weeks**; TBM **4–8 weeks**
DTG with rifampicin	**Twice daily**, until **2 weeks after** rifampicin stops
3HP	Weekly × **12 doses**, **≥2 years**
3HR	Daily × **3 months**, any age
6H	**10 mg/kg** daily × **6 months**
Treatment failure	Positive smear or culture at **month 5** or later
Lost to follow-up	Interruption **≥2 consecutive months**
Progression risk	Infants <1 y: up to **30–40%**
Most disease appears	Within **12 months** of infection

Exam traps

- **"No sputum, so no TB."** Use gastric aspirate, induced sputum, NPA or **stool**; diagnose clinically if needed.
- **"Xpert negative excludes TB."** It doesn't in children.
- **"TST negative excludes TB."** False negatives in HIV, SAM, TBM, miliary TB.
- **"Give isoniazid to every contact."** Only after **excluding active TB** with a symptom screen; isoniazid alone for active TB breeds resistance.
- **"Children need adult mg/kg doses."** They need **higher** mg/kg doses (H 10, R 15).
- **"Keep the same dose for 6 months."** Re-weigh monthly; move up the band.
- **"Start ART on the same day as TBM treatment."** Delay **4–8 weeks** in TBM.
- **"4 months for every child."** Only **non-severe** disease; TBM and bone TB need 12 months.

Questions seniors ask

Viva questions

1. *What are the doses of the first-line drugs in children?* H 10, R 15, Z 35, E 20 mg/kg/day.
2. *What is the Ghon complex?* Primary parenchymal focus plus enlarged draining hilar lymph nodes.
3. *What are the CSF findings in TBM?* Clear, lymphocytes, high protein, low glucose (ratio <0.5), Xpert Ultra may be positive.
4. *Define treatment failure.* Smear or culture positive at month 5 or later.
5. *How does rifampicin interact with ART?* It induces CYP3A4/UGT1A1: double DTG (twice daily), super-boost LPV/r, avoid NVP.
6. *What is non-severe TB (WHO 2022)?* Peripheral node TB, intrathoracic node TB without obstruction, uncomplicated effusion, paucibacillary non-cavitary disease in one lobe, no miliary.

On the ward round

- **What makes you suspect TB in this child?** Contact with an adult with pulmonary TB, cough or fever over 2 weeks, faltering growth, lethargy, and no response to antibiotics or antimalarials.
- **How will you confirm it?** Xpert Ultra on the best available sample (induced sputum, gastric aspirate, NPA or stool; CSF or FNA for EPTB), CXR, HIV test, LF-LAM if HIV-positive and ill; TST as supportive evidence.
- **What regimen and dose would you give a 14 kg child with non-severe PTB?** 2HRZE/2HR: RHZ 75/50/150 three tablets + ethambutol 100 mg three tablets daily for 2 months, then RH 75/50 three tablets daily for 2 months, with pyridoxine; re-weigh monthly.
- **What about the siblings?** Symptom-screen all; evaluate any with symptoms; give TPT to symptom-free contacts (3HR or 6H under 2 y; 3HP or 3HR ≥2 y); follow up monthly; check HIV status.
- **The child is HIV-positive on DTG. What changes?** Start TB treatment, continue ART, give DTG twice daily during rifampicin and for 2 weeks after, cotrimoxazole prophylaxis, watch for IRIS.
- **How do you know TBM, and what is your plan?** Subacute fever, irritability, vomiting, cranial nerve palsy, altered consciousness; CSF lymphocytic with high protein and low glucose; start RHZE (12 months) with prednisolone, manage hydrocephalus and seizures, delay ART 4–8 weeks if HIV-positive.
- **Why is the 4-month regimen safe?** SHINE (NEJM 2022; 1,204 children in Uganda, Zambia, South Africa and India) found 4 months non-inferior to 6 months for non-severe, smear-negative TB (unfavourable outcome 3% in both arms).

Mnemonics

- **Doses "10-15-35-20"**: **H 10, R 15, Z 35, E 20** mg/kg.
- **Weight bands "4-8-12-16-25"**: 1 tab at 4, 2 at 8, 3 at 12, 4 at 16, adult at 25 kg.
- **Suspect TB in a child "C-F-W-L-C"**: **C**ough >2 weeks, **F**ever >2 weeks, **W**eight loss or poor gain, **L**ethargy, **C**ontact.
- **Non-severe TB "PINO"**: **P**eripheral nodes, **I**ntrathoracic nodes without obstruction, uncomplicated **N**on-cavitary disease in one lobe, uncomplicated pleural effusi**O**n.
- **TPT "3-3-6-1"**: **3HP** weekly (≥2 y), **3HR** daily (any age), **6H** daily, **1HP** daily (≥13 y).
- **TB/HIV "Two weeks, but TBM waits"**: ART within 2 weeks; TBM 4–8 weeks.

OSCE tips

- **Counselling a mother starting TB treatment**: what TB is, that it is curable, daily dosing (dispersible in water, same time daily, on an empty stomach if tolerated), **orange urine is normal**, side effects to report (yellow eyes, vomiting, rash), monthly weighing, duration, **screening of all household members**, infection control for the adult source, HIV test.

- **TST reading station**: read induration (not redness) transversely in mm with a ruler at 48–72 h; interpret ≥10 mm (≥5 mm if HIV or SAM).
- **Prescription station**: write drug, FDC strength, number of tablets for the weight band, phase and duration, pyridoxine, and review date.

Presenting this tutorial

Edith presents: Week 3, Tuesday, 08:00–09:30, Dr Mawanda.

Your goal: in about **40 minutes plus 30 minutes of discussion**, leave every classmate able to (1) suspect TB in a child, (2) choose the right sample and test, (3) prescribe by weight band, and (4) manage the household contacts.

Time	Segment	What to say or do
0–3 min	**Hook case**	Kato (section 8): "A 3-year-old with a coughing grandmother and a flat growth chart. Sputum? He can't produce any. What now?" Take 2 answers
3–8 min	**Why children are different**	Burden (1.2 million children a year; most under-fives who die were never diagnosed); progression risk by age; paucibacillary disease; BCG protects against TBM and miliary TB
8–14 min	**Natural history in one diagram**	Draw the Ghon complex → nodes → compression or erosion → haematogenous spread (miliary, TBM) → pleural effusion → bone. Mark the 12-month window
14–22 min	**Diagnosis**	UCG "suspect TB" list; contact history; exam (growth chart!); samples: sputum or induced, gastric aspirate, NPA, **stool**; **Xpert Ultra**; LF-LAM in HIV; CXR features; TST; scores (Edwards ≥7; WHO TDA >10). Stress: **negative Xpert ≠ no TB**
22–32 min	**Treatment**	Doses (10-15-35-20), **the weight-band table**, regimens (4 months non-severe vs 6 vs 12), TBM steroids, monitoring, side effects. One Uganda slide on where the 4-month regimen is given
32–37 min	**TB/HIV and TPT**	ART within 2 weeks (TBM 4–8), DTG twice daily; contact algorithm; 3HP, 3HR, 6H with weight bands; MDR contacts (UCG vs WHO 2024)
37–40 min	**Back to Kato**	Full answer: stool Xpert, regimen, dose, siblings' TPT. Five take-home messages
40–70 min	**Discussion**	Use the questions below; quiz two MCQs

Tips

- **Bring props**: a paediatric RHZ dispersible tablet strip (ask the TB clinic), a growth chart showing faltering, and a printed weight-band table. Show how to dissolve a dispersible tablet in a spoon of water.
- **Make the weight-band table the centrepiece.** Ask a classmate: "This child weighs 13.5 kg; how many tablets?" (3).
- **Name the editions**: "WHO consolidated guidelines Module 5 (2022)", "UCG 2023 section 5.3", "WHO 2024 TPT update". Tutors notice when you quote the current version.
- **Use a CXR image** (hilar nodes, miliary) if you can project; otherwise sketch it.
- **Keep doses exact** and with maxima; **never** say "IPT for everyone" without "after excluding active TB".
- **Rehearse the TBM CSF interpretation**: it is a favourite viva question.
- **Prepare a one-page handout**: suspect list, sample list, weight-band table, regimen table, TPT table.

Questions Dr Mawanda is likely to ask (prepare these)

1. *How do you collect a gastric aspirate?* Fasting overnight (or ≥4 h in infants), early morning before the child gets up, NG tube, aspirate 5–10 mL (or instil 10 mL sterile water and aspirate), send promptly (neutralise if delayed).
2. *Which children get the 4-month regimen?* Non-severe TB (peripheral nodes, uncomplicated intrathoracic nodes, uncomplicated effusion, non-cavitary disease in one lobe), 3 months–16 years.
3. *Why does a negative Xpert not exclude TB?* Paucibacillary disease; Ultra sensitivity in children is only about 60–70% against culture, and most childhood TB is culture-negative too.
4. *When do you start ART in a child with TB?* Within 2 weeks of TB treatment; 4–8 weeks in TBM.
5. *How do you manage a 2-year-old contact with no symptoms?* TPT (3HR or 3HP at ≥2 y, or 6H) after a symptom screen; no TST needed; follow monthly.
6. *What is the steroid dose in TBM?* Prednisolone 2 mg/kg/day (max 60 mg) for 4 weeks, tapered over 2 weeks.

15. References

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3. Turkova A, et al. Shorter Treatment for Nonsevere Tuberculosis in African and Indian Children (SHINE). N Engl J Med. 2022. https://pubmed.ncbi.nlm.nih.gov/35263517/
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7. Medscape (eMedicine). Pediatric Tuberculosis. https://emedicine.medscape.com/article/969401-overview
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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	When should you suspect TB in a child (UCG 2023)?	**Cough >2 weeks**, **fever >2 weeks**, **poor weight gain/weight loss for 1 month**, **reduced playfulness/lethargy/poor feeding** (young children), **close contact** with pulmonary TB; swollen neck/groin nodes.
2	Which children are at highest risk of progressing from TB infection to disease?	**Infants and children <2 (to 5) years**, **HIV-positive**, **severely malnourished**, after **measles**, on steroids. They also get severe forms (**TBM, miliary**). Most disease appears within **12 months** of infection.
3	Samples for Xpert Ultra in a young child who cannot cough up sputum.	**Induced sputum**, **gastric aspirate** (early morning, fasting), **nasopharyngeal aspirate**, **stool** (WHO 2022); for EPTB: CSF, node FNA, pleural fluid. Urine LF-LAM if HIV-positive and seriously ill.
4	How do you collect a gastric aspirate?	Fast overnight (≥4 h in infants); early morning before the child gets up/feeds; pass an NG tube; aspirate 5–10 mL (if none, instil 10 mL sterile water and aspirate); send promptly (neutralise if delayed).
5	Does a negative Xpert exclude TB in a child?	**No.** Childhood TB is paucibacillary; Xpert Ultra detects only about 60–70% of culture-confirmed cases and many children are culture-negative. Diagnose clinically (contact, symptoms, CXR, scores) when needed.
6	Paediatric doses of first-line TB drugs (WHO/UCG).	**Isoniazid 10 (7–15)**, **rifampicin 15 (10–20)**, **pyrazinamide 35 (30–40)**, **ethambutol 20 (15–25)** mg/kg/day. Pyridoxine 25 mg daily (UCG). Memory: **10-15-35-20**.
7	Weight bands for RHZ 75/50/150 and RH 75/50 dispersible FDCs.	**4–7.9 kg: 1 tab; 8–11.9 kg: 2; 12–15.9 kg: 3; 16–24.9 kg: 4; ≥25 kg: adult FDCs.** Ethambutol 100 mg: same number of tablets. Re-weigh monthly.
8	Standard regimen for drug-susceptible TB in children.	**2RHZE/4RH** (6 months) for all forms except TBM and bone/joint TB (UCG 2023).
9	4-month regimen: who qualifies and what is it?	Children **3 months–16 years** (UCG: 2 months–16 y) with **non-severe TB**: **2HRZ(E)/2HR**. Based on the **SHINE** trial (NEJM 2022). UCG 2023: given at general hospital level and above.
10	Define non-severe TB (WHO 2022).	**Peripheral lymph node TB**; **intrathoracic node TB without airway obstruction**; **uncomplicated pleural effusion**; **paucibacillary, non-cavitary disease confined to one lobe, no miliary pattern**.
11	Regimen for TB meningitis and bone/joint TB.	**2RHZE/10RH (12 months)** (UCG 2023). WHO 2022 alternative for TBM: **6HRZ-Eto** (high-dose H and R, Z, ethionamide) for 6 months. Plus steroids in TBM.
12	Steroid regimen in TB meningitis.	**Prednisolone 2 mg/kg/day (max 60 mg) for 4 weeks, then taper over 2 weeks** (UCG: 1–2 mg/kg, max 60 mg). Dexamethasone IV if unable to swallow. Also used in TB pericarditis and airway compression.
13	CSF findings in TB meningitis.	Clear; **lymphocytes 50–500/µL**; **protein high (>1 g/L)**; **glucose low** (CSF:blood <0.5); Gram stain negative; **Xpert Ultra** on CSF may be positive.
14	BMRC staging of TB meningitis.	**I**: GCS 15, no focal signs. **II**: GCS 11–14 or focal signs (e.g. cranial nerve palsy). **III**: GCS ≤10. Outcome worsens with stage.

#	Question	Answer
15	Keith Edwards TB score: threshold and top-scoring items.	**≥7 = treat.** Spine angle deformity 4; illness >4 weeks 3; WFA <60% 3; proven sputum-positive contact 3; positive TST 3; nodes/abdominal mass 3; CNS with abnormal CSF 3; joint/bone swelling 3.
16	WHO 2022 treatment decision algorithms: who and threshold?	Children **<10 years** with presumptive pulmonary TB, at facilities with or without CXR; score symptoms, signs (± CXR); **score >10 → treat for TB**.
17	Tuberculin skin test: how to read and interpret.	0.1 mL (5 TU) PPD intradermally; read **induration** (not redness) transversely at **48–72 h**. Positive **≥10 mm**; **≥5 mm** if HIV-positive or severely malnourished. Shows infection, not disease.
18	When to start ART in a child with TB and HIV.	Start TB treatment first, then **ART within 2 weeks**; **TB meningitis: delay 4–8 weeks** (IRIS risk). Cotrimoxazole prophylaxis for all.
19	Dolutegravir with rifampicin.	Rifampicin induces UGT1A1/CYP3A4 → give **DTG twice daily** (pDTG 10 mg and DTG 50 mg) during rifampicin and for **2 weeks after** it stops. LPV/r: super-boost with ritonavir. Avoid NVP.
20	TB contact screening algorithm for a child.	Symptom screen (**cough, fever, poor weight gain**, lethargy). **Any symptom** → evaluate for TB disease. **None** → TB unlikely → check contraindications → **TPT** (no TST needed in <5 y or HIV).
21	TPT regimens and age limits.	**3HP** weekly × 12 (≥2 y); **3HR** daily × 3 months (any age); **6H** isoniazid 10 mg/kg daily × 6 months (any age); **1HP** daily × 28 days (≥13 y); **6Lfx** for MDR-TB contacts (WHO 2024).
22	3HP child FDC (HP 300/300) weight bands (UCG chart).	**10–15.9 kg: 1 tab; 16–23.9 kg: 1.5; 24–30.9 kg: 2; 31–34.9 kg: 2.5; ≥35 kg: 3** — once weekly for 12 weeks, with pyridoxine.
23	TPT for contacts of MDR-TB: UCG 2023 vs WHO 2024.	UCG 2023: **do not** give TPT to MDR contacts. WHO 2024: **6 months of levofloxacin** after excluding disease. In Uganda: refer/discuss with the MDR-TB centre and follow up for 2 years.
24	Definitions: treatment failure and lost to follow-up.	**Failure**: smear/culture positive at **month 5 or later**. **LTFU**: did not start, or treatment interrupted for **≥2 consecutive months**.
25	Hepatotoxicity on TB treatment: what do you do?	Jaundice, vomiting, abdominal pain or ALT >3× ULN with symptoms (>5× without) → **stop H, R and Z**, check LFTs; when normal, **reintroduce drugs one at a time** (specialist); use a non-hepatotoxic interim regimen if very ill.
26	What does BCG protect against?	Severe childhood TB: **TB meningitis and miliary TB** (~70–80%), much less against pulmonary TB. Given at birth (UNEPI). Not for symptomatic HIV-positive infants (UCG).
27	Why must active TB be excluded before TPT?	TPT (e.g. isoniazid alone) given to a child with **active TB** is **monotherapy** and selects **drug resistance**; the child needs full treatment instead.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 3-year-old boy (11 kg, WAZ −2.4) has had cough, intermittent fever and weight loss for 5 weeks despite a course of amoxicillin. His grandmother, who shares his room, started treatment for Xpert-positive pulmonary TB a month ago. He cannot produce sputum. What is the MOST appropriate next investigation?

A Sputum smear microscopy for AFB
B Tuberculin skin test alone, and treat only if positive
C Repeat amoxicillin for 10 days and review
D Xpert MTB/RIF Ultra on stool or gastric aspirate, with a chest X-ray and HIV test
E Blood culture

2. The 3-year-old in the previous question weighs 11 kg. Stool Xpert Ultra shows 'MTB detected, rifampicin resistance not detected'. CXR shows right hilar lymphadenopathy without airway compression and no cavity. He is HIV-negative with no severe malnutrition. At a general hospital, which regimen is MOST appropriate?

A 2RHZE/10RH for 12 months
B 2HRZE/2HR using RHZ 75/50/150 two tablets + ethambutol 100 mg two tablets daily, then RH 75/50 two tablets daily
C Isoniazid 10 mg/kg daily for 6 months
D 2RHZE/4RH using RHZ 75/50/150 four tablets daily
E Refer to the MDR-TB centre for a second-line regimen

3. A 2-year-old girl (10 kg) has had fever, irritability and vomiting for 12 days. Today she is drowsy (GCS 12) with a left VI nerve palsy. Her father is on TB treatment. CSF: clear, WBC 240/µL (80% lymphocytes), protein 2.0 g/L, glucose 1.3 mmol/L (blood 5.2). What is the MOST appropriate treatment?

A 2RHZE/10RH plus prednisolone 2 mg/kg/day for 4 weeks, then taper
B Ceftriaxone 100 mg/kg/day for 10 days alone
C 2HRZE/2HR plus prednisolone
D Fluconazole 12 mg/kg/day
E 2RHZE/4RH without steroids

4. A 4-year-old HIV-positive boy (15 kg) on ABC/3TC/pDTG is diagnosed with pulmonary TB and started on rifampicin-containing treatment. What change should be made to his ART?

A Stop ART until TB treatment is finished
B Switch to a nevirapine-based regimen
C No change is needed
D Delay ART for 8 weeks
E Give dolutegravir twice daily during rifampicin and for 2 weeks after it stops

5. An 18-month-old girl (10 kg) lives with her mother, who was just diagnosed with Xpert-positive, rifampicin-susceptible pulmonary TB. The child is well, feeding and growing normally, with no cough, fever or weight loss. HIV-negative. What is the MOST appropriate management?

A No action; she is asymptomatic

B Do a TST and give TPT only if ≥10 mm

C Give TB preventive therapy, e.g. 3HR (RH 75/50 two tablets daily for 3 months) or 6H, with monthly follow-up

D Start full treatment with 2RHZE/4RH

E Give 3HP once weekly for 12 weeks

6. A 6-year-old boy (19 kg) is a household contact of his uncle with smear-positive TB. He has had a cough for 3 weeks and has lost weight. What should you do?

A Start 3HP preventive therapy

B Evaluate for TB disease (Xpert on sputum/stool, CXR, HIV test) before deciding

C Give isoniazid preventive therapy for 6 months

D Reassure; children are rarely affected

E Give amoxicillin and repeat the symptom screen in 6 months

7. A 7-year-old girl on 2RHZE for pulmonary TB for 3 weeks presents with vomiting, abdominal pain and yellow eyes. ALT is 8 times the upper limit of normal. What is the MOST appropriate action?

A Continue all drugs and give an antiemetic

B Stop only ethambutol

C Double the pyridoxine dose

D Stop isoniazid, rifampicin and pyrazinamide, monitor LFTs, and reintroduce drugs one at a time when liver tests normalise

E Stop all drugs permanently; the TB is probably cured

8. Which ONE of the following children with drug-susceptible TB is eligible for the 4-month regimen (2HRZ(E)/2HR) according to WHO 2022?

A A 5-year-old with a firm, matted right cervical lymph node, no other disease, HIV-negative

B A 2-year-old with a miliary pattern on chest X-ray

C An 8-year-old with a gibbus at T10

D A 4-year-old with TB meningitis

E A 14-year-old with a large cavity in the right upper lobe and smear-positive sputum

9. An 8-year-old girl (20 kg) has been on 2RHZE/4RH for pulmonary TB. At month 2 her sputum was smear-negative. At month 5 her sputum smear is positive and she is losing weight. Her mother admits doses were often missed. What is the correct interpretation and next step?

A Normal; continue the continuation phase to 6 months

B Relapse; restart 2RHZE/4RH immediately

C Lost to follow-up; trace and restart

D Cured; stop treatment

E Treatment failure; send sputum for Xpert (rifampicin resistance) and culture/DST, address adherence, and refer as indicated

10. A 9-month-old HIV-positive infant (7 kg) is diagnosed with TB meningitis and started on anti-TB treatment and prednisolone. She is not yet on ART. When should ART be started? Advanced

A On the same day as TB treatment
B After completion of 12 months of TB treatment
C Four to eight weeks after starting TB treatment
D Only when the CD4 percentage falls below 15%
E Within 48 hours, before the steroid course ends

11. A child weighs 13.5 kg and starts the intensive phase of TB treatment. How many RHZ 75/50/150 dispersible tablets should she take daily?

A Three tablets
B One tablet
C Two tablets
D Four tablets
E Five tablets

12. A 10-year-old girl (28 kg) has a 6-week history of fever and right-sided chest pain. There is stony dullness and absent breath sounds over the lower half of the right chest. Pleural tap yields straw-coloured fluid: protein 48 g/L, WBC 1,800/µL (90% lymphocytes), Gram stain negative. Xpert on pleural fluid is negative. What is the MOST likely diagnosis?

A Parapneumonic empyema
B Heart failure (transudate)
C Nephrotic syndrome
D Tuberculous pleural effusion
E Pneumococcal pneumonia with resolving effusion

13. A 5-year-old boy (17 kg) is a household contact of his father, who has rifampicin-resistant TB confirmed by Xpert. The boy has no symptoms. According to WHO 2024, which preventive approach is recommended? Advanced

A 3HP once weekly for 12 weeks
B Isoniazid 10 mg/kg daily for 6 months
C 3HR daily for 3 months
D No preventive therapy and no follow-up
E Levofloxacin daily for 6 months, with follow-up for 2 years

14. A 3-year-old HIV-positive boy (12 kg) with advanced HIV is admitted very sick with fever, weight loss and cough. He cannot produce sputum. Which additional rapid test is MOST useful in this situation?

A Erythrocyte sedimentation rate
B Urine lateral flow LAM (LF-LAM)
C Mantoux test read at 72 hours
D Serum CRP
E Widal test

15. A 4-year-old (14 kg) on 2RHZE for 1 month for pulmonary TB is brought for review. His mother is worried because 'his urine has turned orange-red'. He is eating well, gaining weight, and has no jaundice or abdominal pain. What should you tell her?

A Stop rifampicin immediately because of haematuria
B This is a sign of liver failure; admit for LFTs
C Orange-red urine, tears and sweat are a harmless effect of rifampicin; continue treatment
D It is a sign of ethambutol toxicity; stop ethambutol
E It means the TB is resistant; refer to the MDR-TB centre

Answers & explanations

1. D · Xpert MTB/RIF Ultra on stool or gastric aspirate, with a chest X-ray and HIV test

D is right: in a young child who cannot expectorate, **Xpert Ultra on stool (WHO 2022) or gastric aspirate** (or induced sputum/NPA), plus **CXR** and an **HIV test**, is the recommended approach; Xpert also detects rifampicin resistance. A is wrong: he cannot produce sputum, and smear is far less sensitive than Xpert in paucibacillary disease. B is wrong: TST shows infection, not disease, and is often negative in malnourished children; it should not gate treatment. C is wrong: more antibiotics delay diagnosis. E is wrong: blood culture does not diagnose TB.

2. B · 2HRZE/2HR using RHZ 75/50/150 two tablets + ethambutol 100 mg two tablets daily, then RH 75/50 two tablets daily

B is right: intrathoracic lymph node TB without airway obstruction is **non-severe** → the **4-month regimen** 2HRZ(E)/2HR (UCG 2023 at general hospital level and above); at 11 kg he is in the **8–11.9 kg band = 2 tablets**. A is wrong: 12 months is for TBM and bone/joint TB. C is wrong: isoniazid alone is TPT, not treatment, and would select resistance. D is wrong: 6 months is acceptable but 4 tablets is the 16–24.9 kg band (overdose for 11 kg). E is wrong: rifampicin resistance was not detected.

3. A · 2RHZE/10RH plus prednisolone 2 mg/kg/day for 4 weeks, then taper

A is right: subacute course, cranial nerve palsy, reduced consciousness, a TB contact and **lymphocytic CSF with very high protein and low glucose** = **TB meningitis (BMRC stage II)** → **12-month regimen** plus **prednisolone 2 mg/kg/day (max 60 mg)** for 4 weeks then taper. B is wrong: bacterial meningitis is acute with neutrophils; ceftriaxone alone will not treat TBM (it may be given briefly while waiting, but not alone). C is wrong: the 4-month regimen is only for non-severe TB. D is wrong: cryptococcal meningitis is mainly an HIV-associated disease and does not fit this picture without a positive CrAg. E is wrong: TBM needs 12 months and adjunctive steroids, which reduce death and disability.

4. E · Give dolutegravir twice daily during rifampicin and for 2 weeks after it stops

E is right: rifampicin induces DTG metabolism, so **DTG is given twice daily** during rifampicin treatment and for 2 weeks after it ends (Uganda consolidated HIV guidelines, WHO). A is wrong: stopping ART increases mortality. B is wrong: nevirapine has serious interactions with rifampicin and is avoided. C is wrong: without adjustment DTG levels fall, risking failure and resistance. D is wrong: he is already on ART; delaying ART 4–8 weeks applies to starting ART in TB meningitis.

5. C · Give TB preventive therapy, e.g. 3HR (RH 75/50 two tablets daily for 3 months) or 6H, with monthly follow-up

C is right: a **household contact <5 years** with a **negative symptom screen** should receive **TPT**; at 18 months, **3HR** (any age; 2 tablets in the 8–11.9 kg band) or **6H** are suitable. A is wrong: under-fives exposed at home have a high risk of disease. B is wrong: TST is **not required** before TPT in children <5 years (UCG/WHO). D is wrong: there is no evidence of TB disease. E is wrong: 3HP is recommended from **2 years** of age.

6. B · Evaluate for TB disease (Xpert on sputum/stool, CXR, HIV test) before deciding

B is right: a contact with **symptoms** (cough >2 weeks, weight loss) fails the symptom screen and must be **evaluated for TB disease** before any TPT decision. A is wrong: TPT is only for contacts without symptoms; giving it to a child with active TB is inadequate treatment and selects resistance. C is wrong for the same reason. D is wrong: children exposed at home are at high risk. E is wrong: a 6-month delay could allow progression to severe disease.

7. D · Stop isoniazid, rifampicin and pyrazinamide, monitor LFTs, and reintroduce drugs one at a time when liver tests normalise

D is right: symptomatic **drug-induced hepatitis** → stop the hepatotoxic drugs (**H, R, Z**), check LFTs, and **reintroduce sequentially** under specialist guidance (with a non-hepatotoxic interim regimen if she is very ill). A is wrong: continuing can cause fulminant liver failure. B is wrong: ethambutol is not hepatotoxic. C is wrong: pyridoxine prevents neuropathy, not hepatitis. E is wrong: 3 weeks of treatment cannot cure TB; stopping permanently risks relapse and resistance.

8. A · A 5-year-old with a firm, matted right cervical lymph node, no other disease, HIV-negative

A is right: **peripheral lymph node TB** is **non-severe** and qualifies for the 4-month regimen. B is wrong: miliary TB is severe. C is wrong: spinal (osteoarticular) TB needs 12 months. D is wrong: TBM needs 12 months (or the 6-month intensive regimen) plus steroids. E is wrong: cavitary, smear-positive disease is excluded from the non-severe definition.

9. E · Treatment failure; send sputum for Xpert (rifampicin resistance) and culture/DST, address adherence, and refer as indicated

E is right: a **positive smear at month 5 or later = treatment failure** (UCG 2023): send **Xpert** (and Xpert XDR/culture/DST) to look for resistance; if rifampicin-resistant, refer to the MDR-TB centre; if sensitive, restart first-line with adherence support. A is wrong: a positive smear at month 5 is never normal. B is wrong: relapse refers to recurrence after completed and successful treatment. C is wrong: she has not interrupted for ≥2 consecutive months by definition. D is wrong: a positive smear means active disease.

10. C · Four to eight weeks after starting TB treatment

C is right: in **TB meningitis**, ART is **delayed 4–8 weeks** after starting TB treatment to reduce severe neurological IRIS and death (WHO). For other forms of TB, ART is started within 2 weeks. A and E are wrong: very early ART in TBM increases serious adverse events. B is wrong: waiting a year greatly increases HIV-related death. D is wrong: all children with HIV are offered ART regardless of CD4.

11. A · Three tablets

A is right: 13.5 kg falls in the **12–15.9 kg band = 3 tablets** of RHZ 75/50/150 (plus 3 × ethambutol 100 mg). B is wrong: 1 tablet is for 4–7.9 kg. C is wrong: 2 tablets is for 8–11.9 kg. D is wrong: 4 tablets is for 16–24.9 kg. E is wrong: there is no 5-tablet paediatric band; from 25 kg adult formulations are used.

12. D · Tuberculous pleural effusion

D is right: a subacute **lymphocytic exudate** (protein >30 g/L) in a school-age child is **TB pleurisy** until proven otherwise; pleural Xpert has low sensitivity, so a negative result does not exclude it. Treat with 2RHZE/4RH (uncomplicated effusion counts as non-severe for the 4-month regimen). A is wrong: empyema is pus with neutrophils and an acute illness. B and C are wrong: these cause bilateral transudates (low protein). E is wrong: parapneumonic effusions are neutrophilic and follow an acute pneumonia.

13. E · Levofloxacin daily for 6 months, with follow-up for 2 years

E is right: WHO 2024 recommends **6 months of levofloxacin** for household contacts of **MDR/RR-TB** after excluding disease, with clinical follow-up (UCG 2023 still says no TPT; refer to the MDR-TB centre). A and C are wrong: rifamycin-based TPT will not protect against rifampicin-resistant strains. B is wrong: isoniazid is unlikely to work if the strain is MDR (often isoniazid-resistant too). D is wrong: contacts of DR-TB need close follow-up regardless.

14. B · Urine lateral flow LAM (LF-LAM)

B is right: **urine LF-LAM** is recommended for **HIV-positive children who are seriously ill, have advanced HIV, or are inpatients with TB symptoms**; a positive result supports starting treatment immediately, alongside Xpert on any available sample. A is wrong: ESR is non-specific. C is wrong: TST is often negative in advanced HIV and takes 72 hours. D is wrong: CRP is non-specific. E is wrong: Widal is for typhoid and is unreliable.

15. C · Orange-red urine, tears and sweat are a harmless effect of rifampicin; continue treatment

C is right: **rifampicin** colours urine, tears and sweat **orange-red**; it is harmless and should be explained at the start of treatment (it can also stain soft contact lenses). A is wrong: this is not blood. B is wrong: there are no features of hepatitis. D is wrong: ethambutol causes optic neuritis, not urine discolouration. E is wrong: urine colour has nothing to do with resistance.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The grandmother's cough

Kato, a 3-year-old boy from Kisenyi, is brought to Mengo paediatric OPD with a 5-week cough, night sweats and weight loss. He completed amoxicillin 2 weeks ago without improvement. His grandmother, who sleeps in the same room, started TB treatment a month ago. Two siblings, aged 18 months and 6 years, live at home.

1. What features make TB likely, and what else would you ask?

> Cough and night sweats **>2 weeks**, **weight loss**, **no response to antibiotics**, and a **household contact** with pulmonary TB. Ask: grandmother's Xpert result and resistance, other coughing adults, HIV status of the child and mother, BCG, measles, feeding, lethargy, neck swellings, headache/vomiting (TBM).

2. Exam: 11 kg (WAZ −2.4, WHZ −1.8), MUAC 12.8 cm, T 37.9 °C, RR 32, SpO_2 96%, small right cervical nodes, few crackles right upper zone, BCG scar present, no neck stiffness. Which investigations?

> **Xpert MTB/RIF Ultra on stool** and/or **gastric aspirate/induced sputum**; **CXR** (AP + lateral); **HIV test**; consider TST (supportive); FNA of nodes if large. Weigh on a calibrated scale and plot the growth chart.

3. Stool Xpert: MTB detected, rifampicin resistance not detected. CXR: right hilar nodes, small right upper-lobe opacity, no cavity or compression. HIV-negative. Classify and treat.

> **Bacteriologically confirmed, drug-susceptible, non-severe pulmonary TB.** At a general hospital: **2HRZE/2HR**. 11 kg = **8–11.9 kg band**: RHZ 75/50/150 × **2** + ethambutol 100 mg × **2** daily for 2 months, then RH 75/50 × **2** daily for 2 months; pyridoxine; register; treatment supporter.

4. What about his nutrition and follow-up?

> Moderate underweight: nutrition counselling and supplementary feeding (MAM programme/food support); **monthly weight** with band adjustment (move to 3 tablets at 12 kg); check adherence and side effects; expect weight gain within 1–2 months.

5. How do you manage the siblings?

> Symptom-screen both. If well: **18-month-old → 3HR** (RH 75/50 by weight band, daily × 3 months) or 6H; **6-year-old → 3HP** (weekly × 12) or 3HR. No TST needed for the toddler. Any symptoms → full TB evaluation. Check HIV status of the household; follow up monthly.

6. Six weeks later he has gained 1.2 kg. What do you change?

> He now weighs ~12.2 kg → **12–15.9 kg band**: increase to **3 tablets** of RHZ (and 3 of ethambutol), and later RH × 3 in the continuation phase. Continue to the end of month 4; record outcome as **treatment completed** (or cured if a negative follow-up result is available).

Take-away: In childhood TB the contact history and growth chart make the diagnosis, stool or gastric Xpert confirms it, the weight band sets the dose (re-check monthly), and the siblings get TPT.

Case 2: Fever, vomiting and a squint

Esther, a 2-year-old girl (10 kg), is referred from a HC IV in Wakiso with 12 days of fever, irritability and vomiting, treated twice for malaria. Today she is drowsy and her mother noticed a squint. Her father is on TB treatment.

1. What is your leading diagnosis and immediate actions?

> **TB meningitis** (subacute fever, irritability, vomiting, **VI nerve palsy**, drowsiness, TB contact). ABC, glucose, SpO_2, GCS, seizure control; check for signs of raised ICP; plan **LP**; malaria test; HIV test; start empirical ceftriaxone if bacterial meningitis cannot be excluded.

2. GCS 12, left VI nerve palsy, neck stiffness. CSF: clear, WBC 240 (80% lymphocytes), protein 2.0 g/L, glucose 1.3 (blood 5.2), Gram negative, CSF Xpert Ultra 'trace'. Interpret.

> **TB meningitis, BMRC stage II** (GCS 11–14 with focal sign). Lymphocytic CSF, very high protein, CSF:blood glucose 0.25; Xpert 'trace' counts as positive in children.

3. Write the treatment.

> **2RHZE/10RH** (12 months): 10 kg = 8–11.9 band → RHZ 75/50/150 × 2 + ethambutol 100 mg × 2 daily (via NG if needed), then RH × 2; **pyridoxine**; **prednisolone 20 mg daily (2 mg/kg)** for 4 weeks then taper over 2 weeks. (Alternative where available: WHO 6-month 6HRZ-Eto.) Supportive: NG feeds, seizure control, head-up, fluids.

4. On day 5 her head circumference increases and the fontanelle is bulging, with sunsetting eyes. What has happened?

> **Hydrocephalus** (communicating, from basal exudates). Urgent **CT** if available and **neurosurgical review** (VP shunt or endoscopic third ventriculostomy at CURE Children's Hospital Mbale or Mulago); acetazolamide/furosemide may be used as a bridge in communicating hydrocephalus under specialist advice.

5. Her HIV test is positive. When will you start ART, and what else?

> **Delay ART for 4–8 weeks** after starting TB treatment (TBM) to reduce IRIS; start **cotrimoxazole prophylaxis** now; when ART starts with DTG, give **DTG twice daily** while on rifampicin; test and support the mother; screen the household and give TPT to contacts.

> **Take-away:** Subacute fever with a cranial nerve palsy and lymphocytic, high-protein, low-glucose CSF is TB meningitis: 12 months of treatment plus steroids, watch for hydrocephalus, and delay ART 4–8 weeks.

Case 3: TB and HIV in a school-age child

Moses, a 9-year-old boy (22 kg) on ABC/3TC/DTG since age 4, comes to the HIV clinic with 3 weeks of cough, fever and weight loss of 2 kg. His viral load 6 months ago was suppressed. His older cousin, who lives with them, has been coughing for 2 months.

1. How do you evaluate him?

> He screens positive for TB symptoms. **Xpert Ultra on sputum** (he can likely expectorate; induce if not), **CXR**, **urine LF-LAM** if he is seriously ill or has advanced disease, CD4 and viral load, weight/BMI-for-age Z-score. Ask the cousin to be tested (cough triage, Xpert).

2. Sputum Xpert: MTB detected, rifampicin resistance not detected. CXR: right middle-lobe consolidation without cavitation. Which regimen and dose?

> Non-severe disease is possible (one lobe, non-cavitary), but many clinicians give **6 months (2RHZE/4RH)** to HIV-positive children; follow your unit. At 22 kg = **16–24.9 kg band**: RHZ 75/50/150 × **4** + ethambutol 100 mg × **4**, then RH 75/50 × **4**; pyridoxine.

3. What changes to his ART?

> Continue ART; give **DTG 50 mg twice daily** while on rifampicin and for **2 weeks after** rifampicin stops; continue ABC/3TC; **cotrimoxazole prophylaxis**; repeat viral load as scheduled.

4. Three weeks later he has a new fever and enlarging neck nodes, though adherent. What do you consider?

> **Paradoxical TB-IRIS** is less likely as he was already on ART, but possible; also consider **drug resistance** (check Xpert/DST), another infection, poor drug absorption, or lymphoma. Review adherence, re-test, and seek specialist advice; continue TB treatment and ART unless otherwise directed.

5. What happens to the household?

> The cousin is evaluated and, if TB is confirmed, treated (likely the source). All household contacts are symptom-screened; **TPT** for well children <5 and HIV-positive contacts (and older contacts per programme); infection control at home (ventilation, cough etiquette).

Take-away: In a child with HIV, screen for TB at every visit; with rifampicin give DTG twice daily, keep cotrimoxazole, and track down the coughing adult at home.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.